DataAI - Multidimensional Balancing

Business guide for balancing, comparing, and interpreting changing matrices and multidimensional totals

Important distinction: this guide separates classic matrix balancing from true multidimensional balancing. Scenarios 1a-3c balance two-dimensional row/column matrices or wide matrices where selected fields act as columns. Only scenarios 4a, 4b, and 4c are true multidimensional balancing, because they balance totals across multiple selected dimensions instead of only one row field and one column field.

Multidimensional balancing is used when a starting data structure must be adjusted so its selected totals match a target structure while preserving as much of the starting pattern as possible. In business terms, it answers: if totals changed across several dimensions, what cell-level changes are consistent with the original data and the requested target totals?

What The Method Does

The method begins with the current report data chosen by the user. The user selects categories, measures, aggregation options, scenario, iteration settings, and optional manual target sums. DataAI converts those selections into one or more matrices or multidimensional summary tables, balances the starting structure to the target totals, and presents the resulting matrices, coefficients, comparisons, exports, and AI-ready interpretation output.

• Starting Matrix: the original matrix or multidimensional table built from the selected report data. In scenarios 1a-3c it is mainly a row/column matrix. In scenarios 4a-4c it is built by selected dimensions.

• Target Matrix or Target Sums: the requested or observed target structure. It may be entered manually, derived from a second measure, or derived from the same measure under another period, scenario, status, or other comparison value.

• Balancing Matrix: the adjusted structure. It has totals close to the target totals while keeping the original proportions as much as the iterative method allows.

• Balancing Coefficients: multipliers that explain how strongly each row, column, or selected dimension had to change to fit the target.

• Precision and Number of Steps: controls that define how close the result must be to the target and how many iterations may be attempted.

Scenario Families

Scenario family	Business purpose	Type
1a-1b	Balance a starting matrix to manually entered row and column totals.	Classic two-dimensional matrix balancing
2a-2c	Balance one measure against another measure or against another value of the same comparison field.	Comparison matrix balancing
3a-3c	Repeat balancing across several selected measures or wide matrix columns.	Wide/multi-column matrix balancing
4a-4c	Balance totals across selected dimensions, such as region, product, channel, period, or scenario columns.	True multidimensional balancing

The difference matters because two-dimensional balancing explains movement across a row-by-column table, while multidimensional balancing explains movement across several dimensions at the same time. The 4a-4c scenarios are the ones to use when the business question is about many dimensions shaping the same total.

Main Inputs

Input	Meaning	Typical use
Scenario	Defines whether the work is manual total balancing, target matrix balancing, iteration balancing, or true multidimensional balancing.	Choose 4a, 4b, or 4c for multidimensional balancing.

Input	Meaning	Typical use
Row field	The main category used as matrix rows.	Region, department, customer group, product family, location.
Column field	The main category used as matrix columns.	Channel, product, period, status, scenario.
Measure field	The numeric value being balanced.	Sales, quantity, cost, revenue, claims, inventory, units.
Second measure or comparison value	The target or comparison value used in scenarios where one structure is balanced against another.	Compare one year to another, plan to actual, base scenario to target scenario.
Selected columns / dimensions	Additional fields used as separate balancing dimensions or multiple target measures.	Used most importantly in 4a-4c.
Manual row and column sums	User-entered target totals when no target matrix is supplied.	Budget targets, planned totals, regulatory totals, expected totals.
Number of steps	Maximum number of balancing iterations.	Higher values allow the result more chances to reach tight precision.
Precision	Allowed mismatch between balanced totals and target totals.	Smaller values require closer matching.

Algorithm In Plain Language

The balancing engine uses an iterative proportional fitting approach, often known in analytics as RAS-style balancing. It repeatedly adjusts rows, columns, or selected dimensions so that the starting structure moves toward the target totals without throwing away the original pattern.

• Step 1: build the starting matrix or multidimensional table from selected fields and aggregation options.

• Step 2: build or read the target totals. Targets can come from manual totals, another measure, another period, or selected dimensions.

• Step 3: when needed, scale starting and target totals to the same overall basis so the comparison is meaningful.

• Step 4: apply balancing multipliers to move the starting structure toward the target.

• Step 5: repeat until the mismatch is within the selected precision or the maximum number of steps is reached.

• Step 6: show the final balanced matrix, coefficients, differences, and records that explain where the most important changes happened.

Sample: Comparing Two Matrices

This small example shows a classic two-dimensional matrix balancing case. It is not the true multidimensional 4a-4c case, but it demonstrates the same balancing idea in the simplest possible form.

Region	Channel	Sales
North	Online	120
North	Retail	80
South	Online	90
South	Retail	110

Starting matrix

Region	Online	Retail	Total
North	120	80	200
South	90	110	200
Total	210	190	400

Target totals

Target row totals	North = 220, South = 180
Target column totals	Online = 230, Retail = 170
Grand total	400

Balanced result

Region	Online	Retail	Total
North	141.11	78.89	220.00
South	88.89	91.11	180.00
Total	230.00	170.00	400.00

The output shows that North Online had to rise strongly, South Retail had to fall, and the final matrix now matches both row and column target totals. In a true multidimensional case, the same principle is applied across selected dimensions such as Region, Product, Channel, Period, and Scenario.

How To Read The Results

• A large positive change in a cell means that part of the business had to grow more than the original pattern suggested.

• A large negative change means that cell had to shrink or contribute less to match the target structure.

• A coefficient above 1 means an area was scaled up; below 1 means it was scaled down.

• When the same row, column, or dimension repeatedly receives large coefficients, it may be driving the overall change.

• Differences between starting and balanced values help identify which changes matter most, not just which totals changed.

Using Balancing To Compare Two Matrices

Matrix balancing can compare two matrices against each other even when every number is different. For example, the starting matrix may represent one year and the target matrix may represent another year, several years, a plan, a forecast, or a new market scenario. The question is not simply which cells changed. The stronger question is: which cell changes actually explain the development of the whole structure?

> The method gives a metric-driven way to evaluate trends in cells and compare complex data development. Instead of looking at isolated row totals or column totals, the user can see which cells, groups, dimensions, or coefficients most affect the final balanced output.

• Year-to-year comparison: balance last year to this year and identify which cells explain the structural shift.

• Plan versus actual: balance planned values to actual totals and see where the plan was most unrealistic.

• Scenario comparison: compare base, optimistic, and pessimistic scenarios using the same structure.

• Multi-year development: use iteration scenarios to see how a structure moves from a starting value to a target value.

• Complex data development: use 4a-4c when multiple selected dimensions must be evaluated together.

How True Multidimensional Scenarios 4a-4c Are Different

Scenario	What it answers	When to use it
4a	How should a starting measure be balanced so its selected-dimensional totals match a target measure?	Use when both starting and target measures exist and dimensions must be matched.
4b	How does the starting structure for one comparison value move toward the target structure for another comparison value?	Use for period-to-period, scenario-to-scenario, or status-to-status development.
4c	How should a starting measure be balanced to manually entered selected-dimensional totals?	Use when target totals come from planning, policy, budget, or outside assumptions.

In 4a-4c, selected dimensions are not just decorative fields. They define the totals that the balancing process must satisfy. This is why these are the true multidimensional scenarios.

Recommended Interpretation Workflow

• Start with a simple matrix view to understand the basic row and column pattern.

• Choose a comparison target: another year, another scenario, another measure, or manually entered totals.

• Run the balancing with reasonable precision and enough steps.

• Review the balanced matrix and coefficients together. The matrix shows where values landed; coefficients show how strongly values had to move.

• Use difference views to find cells where the balanced result differs most from the starting or target structure.

• Use AI interpretation after the result is built to summarize which cells, dimensions, and coefficients explain the change.

Business Uses

Use case	How balancing helps
Sales and revenue planning	Shows how sales by region, product, channel, or period would need to move to meet target totals.
Budget allocation	Adjusts department or project allocations to match approved totals while preserving the original allocation pattern.
Market development	Compares market structures across years or scenarios to see what changed in the strongest cells.
Inventory and supply planning	Balances movement by product, location, or period against supply targets.
Risk and claims analysis	Compares claim or risk matrices across periods and identifies where structural changes are concentrated.
Operational performance	Evaluates which combinations of department, location, category, or time period explain performance movement.
Complex data development	Provides a practical way to compare multidimensional data structures and understand which cells affect the output.

What To Export Or Ask AI

- Export the starting matrix to document the original structure.
- Export the target matrix or target sums to show the comparison basis.
- Export the balancing matrix to show the adjusted result.
- Export coefficients to explain the scale and direction of changes.
- Export difference tables to show which cells most affect the output.
- Ask AI to explain the largest changes, strongest coefficients, outliers, and business meaning of the balanced result.

Checklist For A Meaningful Balancing Analysis

- The starting matrix and target matrix represent comparable business structures.
- Row, column, and selected-dimension fields are real business categories, not technical identifiers.
- The measure field is numeric and meaningful for balancing.
- The target totals make business sense and are not accidental duplicates or incomplete totals.
- Precision is tight enough to trust the result but not so tight that the process runs unnecessarily long.
- For multidimensional analysis, scenario 4a, 4b, or 4c is used.
- The final interpretation includes both balanced values and balancing coefficients.

End of Version 2 business guide. This version intentionally avoids programming, page, and internal implementation references.